AI Teaching Assistant for Personalized Learning Support using LLM and RAG

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***Abstract*—As we converge into the age of Artificial Intelligence (AI), not only does AI have the potential to upend traditional approaches to teaching and learning, but it can facilitate personalized, situational and efficient solutions as well. For this project, we will use the LLMs and Retrieval-Augmented Generation RAG to build an AI Teaching Assistant. Using the textbook, academic resources, and cuts from a MySQL database containing performance records, the assistant categorizes student inquiries, showcases useful information from the content, and creates accurate and concise answers.**

***Index Terms*—Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Personalized Learning, Context-Aware Responses, Student Performance Analysis, Query Classification, Knowledge Retrieval, Educational Support, Subject-Specific Assistance, Data-Driven Learning, Study Recommendations**

I. Introduction

Education is one of the key frameworks for any society and we have seen that currently it is significantly transforming a lot today because of the revolution of technology that is why we have observed all the challenges which are to adopt the education system to meet the organizational need of the students. The technology employed in the Teaching Assistant project includes LLMs, and the RAG techniques that help deliver an excellent teaching approach based on a student's individuality. Where the conventional approaches may sometimes not target individuals' learning modalities and rates, this system targets individual learning rates in real-time.

The major goal therefore is to provide accurate and relevant answers to questions most students are likely to ask in line with the study material. As a result, the assistant answers not only in an academic way, but also finds ways of explaining the information in a textbook that can easily be comprehended. Further, it collects interaction data to identify learning related issues in order for a teacher to enhance his/her teaching methods to assist a learner.

This approach benefits learners because they get immediate individual feedback as well as benefit educators who get useful information about students' performance. This project therefore dreams of a society where technology will supplement education to have intelligent tools complement intelligent teachers in the learning process.

II. Flowchart

The flowchart below explains the entire process starting from student asking a query and getting response from the LLM. The heart of the entire system lies in the resources which include textbooks. First, we divided the textbook into chunks and then stored into vector embeddings. This embeddings capture the the semantic meaning of the textbook data present.

We have student data which includes the grade of student across all the subjects and courses he/she registered in current study year. It also contains the queries that student asked the chat bot regarding the topics which the student is not performing well. This data is stored in MYSQL database. The LLM has access to both the vector data base and the MYSQL database

At last, as soon as the user enters a query it will be matched with relevant content from the vector database and now LLM gets the information and outputs according to the query with the knowledge of the textbooks.

We also built a user interface for easy navigation access to the chat bot and helps the student to use without worrying about the technical details

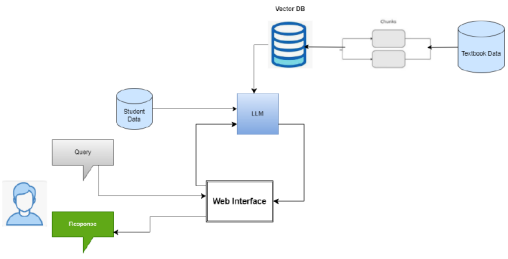

Figure 1: System Architecture

III. Literature of Review

A. *Scaffolding and Personalization in AI Teaching Assistants*

Concerns such as scalability and the ability to provide feedback, while keeping both time and individual students in context, are met by the use of Teaching Assistants (TAs) that are designed and built based on Artificial Intelligence techniques. Modern AI systems utilize Large Language Models

(LLMs) to perform in roles consisting of debugging, code generation as well as conceptual direction. Unlike traditional human TAs that may promote solution-seeking teaching measures, utilization of the AI systems promotes critical thinking and self-reliance from a clearly sequenced feedback system. This approach allows students solve the problems on their own while at the same time encouraging them develop better understanding. The study emphasizes that scaffolding is a major component of the design proposition that allows the students to interact with such tasks without getting overwhelmed (Denny et al., 2024). Additional customization according to the experience of learners also improves the work done, reducing the cognitive load even more and offering what is specifically relevant to students: Beginning students often need more structure and guidance, while experienced students may require just occasional suggestions and reminders throughout the learning process (Savelka et al., 2023). These artificial intelligence integrated systems are compatible with pedagogical objectives, as well as, engage students in the thinking processes and enable them to make decisions. Their efficacy makes them important resources in contemporary learning.

B. *The Role of AI Agents in Personalized Education*

The evidence has also established that AI agents can be integrated into learning environments in an effective manner and change the ways of traditional learning by providing individualized, context-adaptive and accessible approaches. Studied show that avenues such as Large Language Models (LLMs) and virtual agents are affordable and more efficient than human trainers. These systems can provide on-the-spot, context aware help and can solve other real issues such as expensive production in video instruction and non-availability of a human tutor at times of need (Denny et al, 2024).

Moreover, it has been found that conversational and virtual agents are functional in enhancing the access of learning, especially for the disadvantaged learners, by providing impartial standard instructional signals (White et al., 2014). However, to lock and load the students' interest and knowledge retention then the full design must be regarded as powerfully simple, clear, and relevant to the learner's level. In the instances whereby the applications of AI agents have been carried out appropriately, they afford a strong avenue to erase disparities in the availability of education and improve the training process.

C. *Chatbot-based Learning Systems*

The application of chatbot-based systems in learning contexts has recently attracted much interest—and particularly in helping new learners in programmable courses. The use of conversational agents, like the "Coding Tutor", is argued to be useful in mitigaging resources in large scale educational practices according to the study. These systems use features of enhanced nLP and AI to mimic teaching assistants and provide a series of instructions, answer questions and instantly grade assignments. It has been established that they facilitate interactivity, and the creation of individualized learning schemes, which is in line with the engagement levels adjudged key by theories such as the ICAP framework. Chatbots' flexibility, especially in terms of choreographing an approach according to the knowledge background of the students, is another advantage. However, there is a remained limitation of being responsive to chatbot-related issues and expanding the knowledge base As the result of the evaluation, it has been reported that effectively designed chatbot system can enhance the accessibility, reduce the workload of educators, and increase the satisfaction level and learning achievement of students.

IV. Objectives of the Study

The objectives of this study are as follows:

- **Developing AI Teaching Assistant**
  - The main objective is create an interactive chat bot designed especially to assist students with subject-specific queries (e.g., Arts, Mathematics, Science).
  - We have also enabled both text and voice input for processing the queries voices.
- **Implement Personalized AI specific to student**
  - To Use generative AI models to classify, process and reply to queries asked by th student in the context of relevance to subject classify, process, and respond to queries based on context and relevanc if subject matter.
  - To provide good response which is modified according to each student
- **Facilitate Student Performance Analysis:**
  - To make a clear and neat dashboard for analysing the student performance.
  - Generate personalized grade summaries and insights using machine learning and generative AI.
- **Integrate Learning Resources with Semantic Understanding:**
  - To create semantic embeddings from the textbook.
  - To match student queries according to the textbook needs.
- **To add Authentication:**
  - To implement authentication to ensure secure access to the system.
  - To keep track of all queries.

V. Data Collection

The Data set comprises of textbooks collected from various domains including Arts,Mathematics and Science from the online resources. The book also contains feedback questions which has questions in the each lesson.

These textbooks are extracted and converted into transformer embeddings so that it can capture the semantic meaning of the words. .

We have also used synthetic data by collecting students data, courses data, query data. Added some subject data and query history which is later used for the analysis.

VI. Exploratory Data Analysis

A. *Word Cloud Analysis*

The word cloud shown below is generated based on the queries asked by students. It highlights the most frequently mentioned topics, enabling educators to identify common areas of concern and focus their efforts on addressing these issues. Larger words represent more frequent queries, providing insights into the key challenges students face.

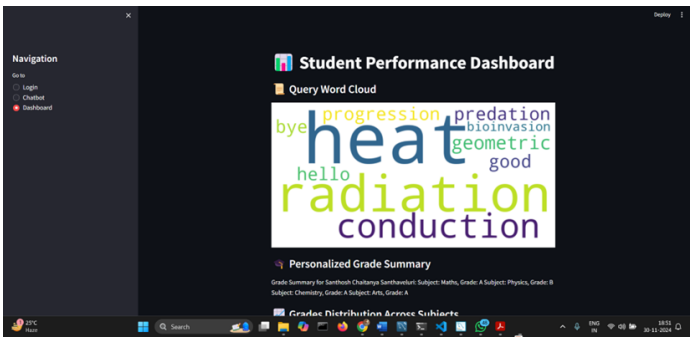

Fig. 1. Word Cloud of Student Queries. Larger words indicate more frequently asked terms, revealing students' main areas of concern such as "heat," "radiation," and "conduction."

In the above image, we can see the topics like heat, radiation, and conduction, which are frequently mentioned in student queries. This indicates that specific students might not be understanding the heat concepts properly, and now teachers can help them by taking additional classes related to the specific subject.

B. *Grade Distribution Analysis*

The grade distribution plot signifies the subjects and grade count for all the students. Many students are finding it difficult to perform well in Chemistry. The graph illustrates how students are performing across all the subjects and identifies areas where they are facing difficulties, enabling teachers to take the required measures to support them.

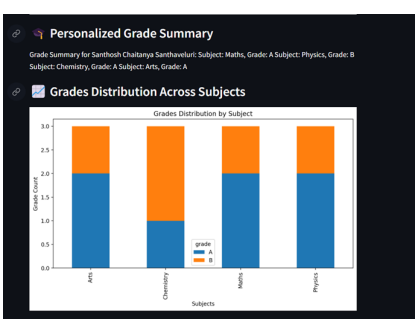

Fig. 2. Grades Distribution Across Subjects.

From the above graph, we can infer:

- The performance of students is almost equal in Maths and Physics.
- The highest grade count is in Arts, while Chemistry has the lowest grade count.
- This indicates that students need to focus on improving their grades in Chemistry to enhance their overall performance.

VII. Data Analytics

A. *Individual Performance Analysis*

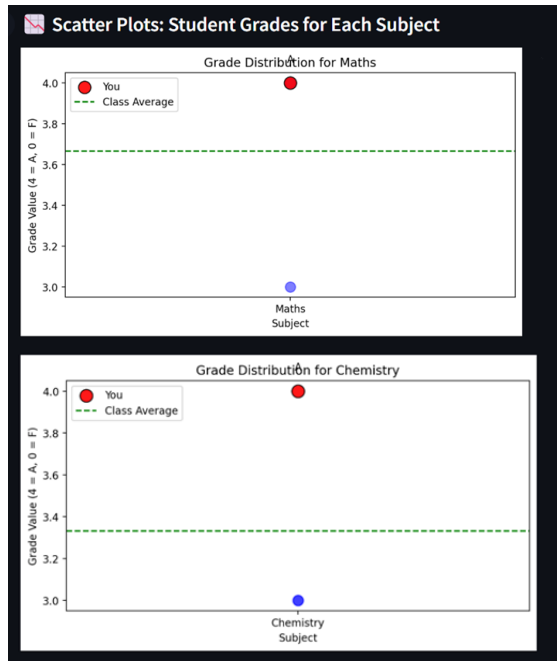

Fig. 3. Scatter plots showing individual performance against class averages.

In the above scatter plots, it tells each individual how they are performing in their own class with regards to each subject. The dotted line indicates the class average, whereas the red dot indicates the individual performance. If it is above the dotted line, the student is performing well, whereas below the line indicates room for improvement in the subject.

With the above plots, students can compare their overall performance with regards to the class.

In the above example, the student performed well in Maths and Chemistry because, in the plots, their scores stand out above the average line. However, in Physics, there is a scope for improvement because their score lies below the class average.

VIII. Data Visualisation and Results Report

The features of the AI Teaching Assistant Chatbot are a voice-enabled interaction system to improve the users' accessibility and engagement. Some of the interaction methods allowed by the Learning Management System include voice input where a student may post a question, which is transcribed and presented to the chatbot. For example, in the provided demonstration: At one time a student typed: "What is conduction?" Voice query was also used to input data into the system and the system produced a clear and accurate transcription for the student to check. Also, the chatbot is able to respond to questions within the domain of interest.

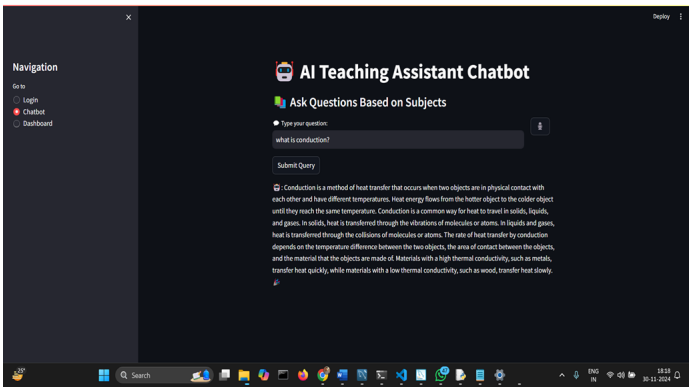

Fig. 4. Voice Interaction and Domain Specific Query Response Demonstration.

- The chatbot searched for information in the information resources available to it and in textbooks.
- The response predicted and described both positive and negative effects, as well as both natural and artificial sources of radiation and what is known as ionizing and non-ionizing radiation.
- Provide answers based on textbook materials in order to remain applicable and not to overload the audience.
- Respond clearly and consistently with educational content for learning objectives in the report.

The technologies used are harmoniously implemented into one single platform through the AI Teaching Assistant. Thus, with the help of such features as domain-specific query classification, voice interaction, as well as the progress bar, the system provides academic assistance as well as performance tracking. The general design and textbook orientation make it advantageous and versatile in classroom use.

IX. Conclusion

The AI Teaching Assistant project demonstrates how modern AI technologies, such as LLMs and RAG, can effectively transform traditional education by offering personalized, accessible, and context-aware learning support. By integrating textbook-based semantic embeddings, voice-enabled interaction, and performance tracking, the assistant provides tailored responses to student queries, enhancing engagement and addressing individual learning needs. The use of domain-specific query classification ensures that responses are both accurate and relevant, making the system highly applicable to diverse subjects and learning contexts.

This innovative approach benefits both students and educators. Students receive immediate feedback and actionable insights to improve their understanding, while educators gain access to valuable data on student performance, enabling them to refine their teaching strategies. The scalable and versatile design makes the AI Teaching Assistant a valuable tool in modern education, bridging gaps in traditional methods and fostering an adaptive, efficient, and inclusive learning environment.

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